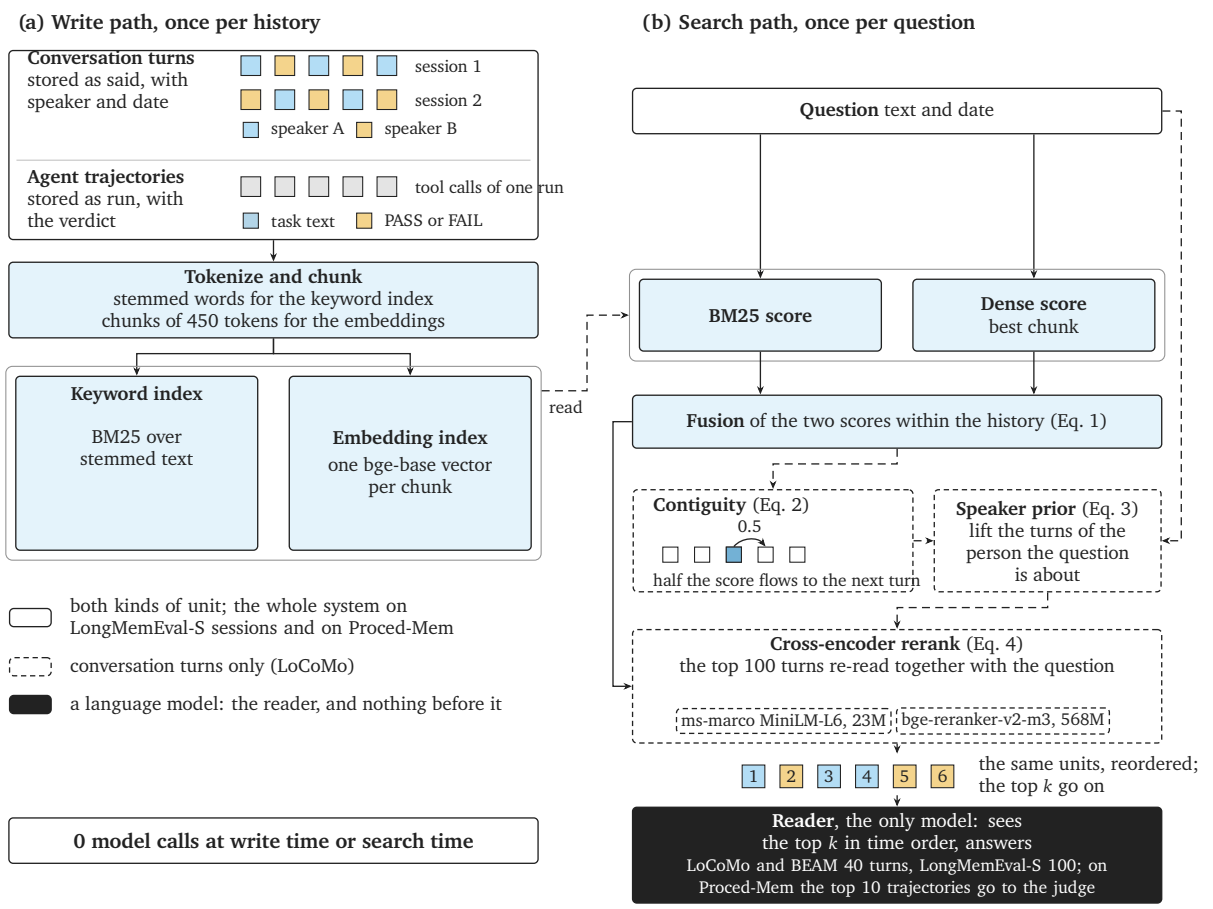
(a) Write path, once per history
Conversation turns
stored as said, with speaker and date
session 1
session 2
speaker A
speaker B
Agent trajectories
stored as run, with the verdict
tool calls of one run
task text
PASS or FAIL
Tokenize and chunk
stemmed words for the keyword index
chunks of 450 tokens for the embeddings
Keyword index
BM25 over stemmed text
Embedding index
one bge-base vector per chunk
read
both kinds of unit; the whole system on LongMemEval-S sessions and on Proced-Mem
conversation turns only (LoCoMo)
a language model: the reader, and nothing before it
0 model calls at write time or search time
(b) Search path, once per question
Question text and date
BM25 score
Dense score
best chunk
Fusion of the two scores within the history (Eq. 1)
Contiguity (Eq. 2)
0.5
half the score flows to the next turn
Speaker prior (Eq. 3)
lift the turns of the person the question is about
Cross-encoder rerank (Eq. 4)
the top 100 turns re-read together with the question
ms-marco MiniLM-L6, 23M
bge-reranker-v2-m3, 568M
1 2 3 4 5 6
the same units, reordered; the top k go on
Reader, the only model: sees the top k in time order, answers
LoCoMo and BEAM 40 turns, LongMemEval-S 100; on Proced-Mem the top 10 trajectories go to the judge